

4.10.9

EE25BTECH11009 - Anshu Kumar Ram

Question:

Find the equation of the plane passing through the line of intersection of the planes

$$\mathbf{r} \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) - 4 = 0, \quad (0.1)$$

$$\mathbf{r} \cdot (2\hat{i} + \hat{j} - \hat{k}) + 5 = 0 \quad (0.2)$$

and which is perpendicular to the plane

$$\mathbf{r} \cdot (5\hat{i} + 3\hat{j} - 6\hat{k}) + 8 = 0. \quad (0.3)$$

Solution:

Given - the equation of the plane passing through the line of intersection of the planes

$$\mathbf{r} \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = 4 \quad (0.4)$$

and

$$\mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + 5 = 0 \quad (0.5)$$

and perpendicular to the plane

$$\mathbf{r} \cdot \begin{pmatrix} 5 \\ 3 \\ -6 \end{pmatrix} + 8 = 0 \quad (0.6)$$

The required plane can be written as a linear combination of the first two planes:

$$(\mathbf{n}_1 + \lambda \mathbf{n}_2)^T \mathbf{r} = c_1 + \lambda c_2 \quad (0.7)$$

where

$$\mathbf{n}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad c_1 = 4 \quad (0.8)$$

$$\mathbf{n}_2 = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}, \quad c_2 = -5 \quad (0.9)$$

Thus, the normal vector of the required plane is

$$\mathbf{n} = \mathbf{n}_1 + \lambda \mathbf{n}_2 \quad (0.10)$$

Since the plane is perpendicular to

$$\mathbf{r} \cdot \begin{pmatrix} 5 \\ 3 \\ -6 \end{pmatrix} + 8 = 0 \quad (0.11)$$

we must have

$$\mathbf{n}^T \mathbf{n}_3 = 0, \quad \mathbf{n}_3 = \begin{pmatrix} 5 \\ 3 \\ -6 \end{pmatrix} \quad (0.12)$$

That is,

$$(\mathbf{n}_1 + \lambda \mathbf{n}_2)^T \mathbf{n}_3 = 0 \quad (0.13)$$

Substituting values,

$$\begin{pmatrix} 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \\ -6 \end{pmatrix} + \lambda \begin{pmatrix} 2 & 1 & -1 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \\ -6 \end{pmatrix} = 0 \quad (0.14)$$

$$-7 + 19\lambda = 0 \implies \lambda = \frac{7}{19} \quad (0.15)$$

Thus,

$$\mathbf{n} = \mathbf{n}_1 + \frac{7}{19} \mathbf{n}_2 = \frac{1}{19} \begin{pmatrix} 33 \\ 45 \\ 50 \end{pmatrix} \quad (0.16)$$

Also,

$$c = c_1 + \lambda c_2 = 4 + \frac{7}{19}(-5) = \frac{41}{19} \quad (0.17)$$

Multiplying through by 19, the required plane is

$$\begin{pmatrix} 33 & 45 & 50 \end{pmatrix} \mathbf{r} = 41 \quad (0.18)$$

Final Answer:

$$33x + 45y + 50z = 41 \quad (0.19)$$

